

Finger print

Title: Secure fingerprint matching system

Topics:

- ✓ python image processing, fingerprint matching algorithm, if-else logic, error handling, multithreading, FIFO/LIFO-data structures, SQL --DDL/DML, python- SQL integration, database query& security

Problem Statement: My scenario is I would like to print a green button if my

thumb is available in the database are print red button

Role/Responsibilities:

Python code:

o Syntax:

```
Thumb-id=input("place your thumb(enter your thumb-id:)")
```

```
Database=["thumb-1","thumb-2",thumb-3"]
```

If thumbed in database:

```
Print("green button: your thumb is available")
```

Else

Print("red button: your thumb is not available")

o Real time code:

Fingerprint database creation

```
fingerprint_db = ["FP101", "FP102", "FP103"]
```

```
# Database to store chesina fingerprints list
```

```
def authenticate_fingerprint(fp_id):
```

```
    # Fingerprint authenticate cheyyadaniki function
```

```
    try:
```

```
        # Error handle cheyyadaniki try block
```

```
            print("Scanning Fingerprint...")
```

```
        # Fingerprint scan start message
```

```
            if fp_id in fingerprint_db:
```

```
                # Fingerprint database to check
```

```
                    print("Fingerprint Found in Database")
```

```
                # Fingerprint match ayyindi
```

```
                    print("ACCESS GRANTED ")
```

```
                # Green signal – access allow
```

```
            else:
```

```
                # Fingerprint match kakapothe
```

```
print("Fingerprint NOT Found in Database")

# Fingerprint dorakaledu

print("ACCESS DENIED ")

# Red signal – access deny


except Exception as e:

# Runtime error vasthe

print("System Error:", e)

# Error message display

# ===== Test Scenarios
=====

authenticate_fingerprint("FP101")

# Valid fingerprint test case

print("-----")

# Output separation line


authenticate_fingerprint("FP999")

# Invalid
```

fingerprint test case

o Output:

Scanning Fingerprint...

Fingerprint Found in Database

ACCESS GRANTED

Scanning Fingerprint...

Fingerprint NOT Found in Database

ACCESS DENIED

Diagram:



Interview Questions:

- ✓ Explain your finger print project in 3 lines
 - o Authentication
- ✓ What is fingerprint matching logic
 - o Comparison
- ✓ Which python/SQL topics did you use
 - o Integration
- ✓ Why use multithreading in this project

- o Performance
 - ✓ What is FIFO/LIFO use here
- o Sequencing
 - ✓ How did you handle errors
- o Exception
 - ✓ Purpose of green/red button
- o Authorization
 - ✓ What database is used
- o SQL